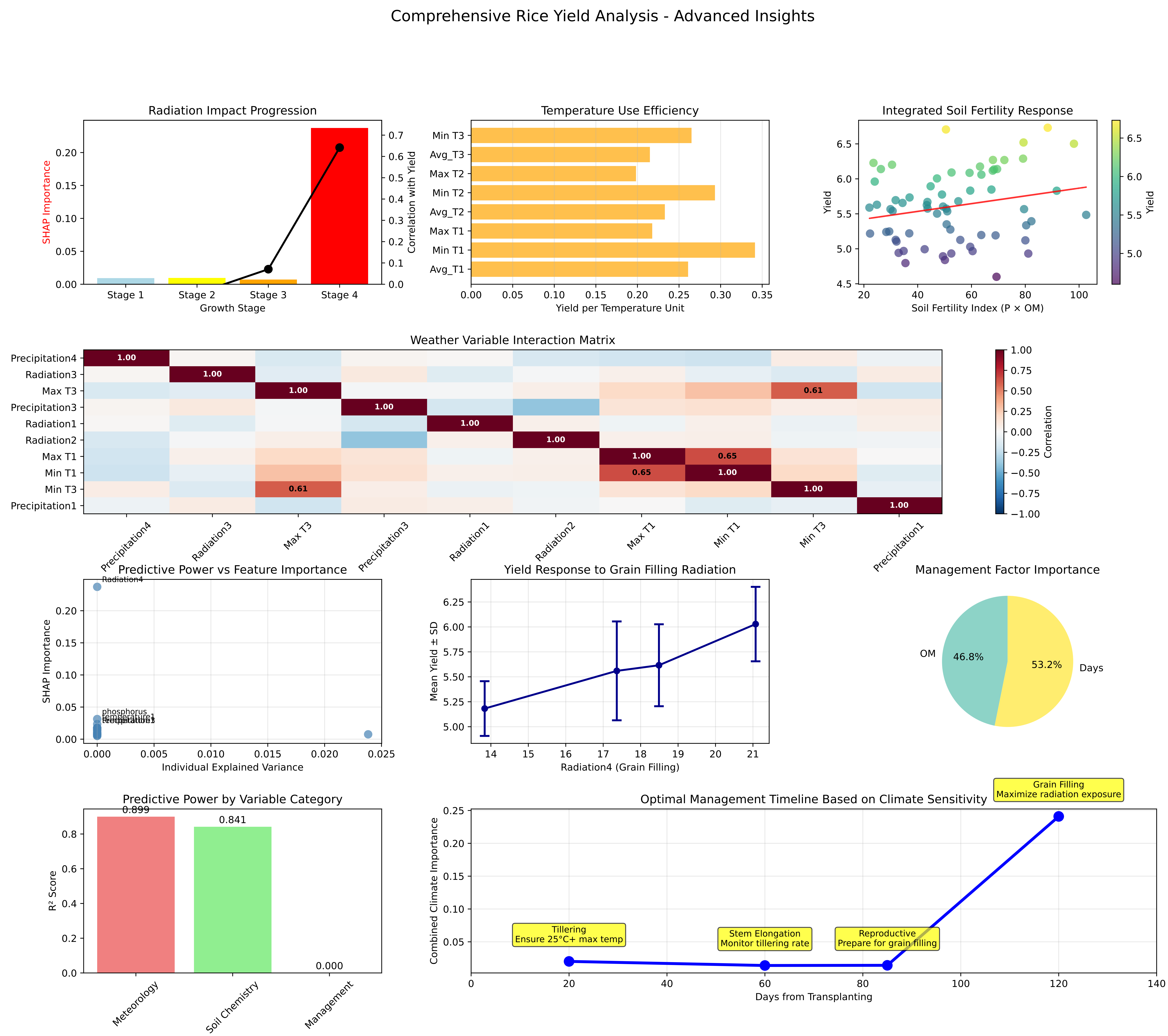


Comprehensive Rice Yield Analysis - Advanced Insights
Radiation Impact Progression
SHAP Importance
Correlation with Yield
Stage 1
Stage 2
Stage 3
Stage 4
Growth Stage
Temperature Use Efficiency
Min T3
Avg_T3
Max T2
Min T2
Avg_T2
Max T1
Min T1
Avg_T1
Yield per Temperature Unit
Integrated Soil Fertility Response
Yield
Soil Fertility Index (P × OM)
Weather Variable Interaction Matrix
Precipitation4
Radiation3
Max T3
Precipitation3
Radiation1
Radiation2
Max T1
Min T1
Min T3
Precipitation1
0.61
0.65
Correlation
Predictive Power vs Feature Importance
Radiation4
phosphorus
SHAP Importance
Individual Explained Variance
Yield Response to Grain Filling Radiation
Mean Yield ± SD
Radiation4 (Grain Filling)
Management Factor Importance
OM
46.8%
53.2%
Days
Predictive Power by Variable Category
0.899
0.841
0.000
R² Score
Meteorology
Soil Chemistry
Management
Optimal Management Timeline Based on Climate Sensitivity
Grain Filling
Maximize radiation exposure
Tillering
Ensure 25°C+ max temp
Stem Elongation
Monitor tillering rate
Reproductive
Prepare for grain filling
Combined Climate Importance
Days from Transplanting

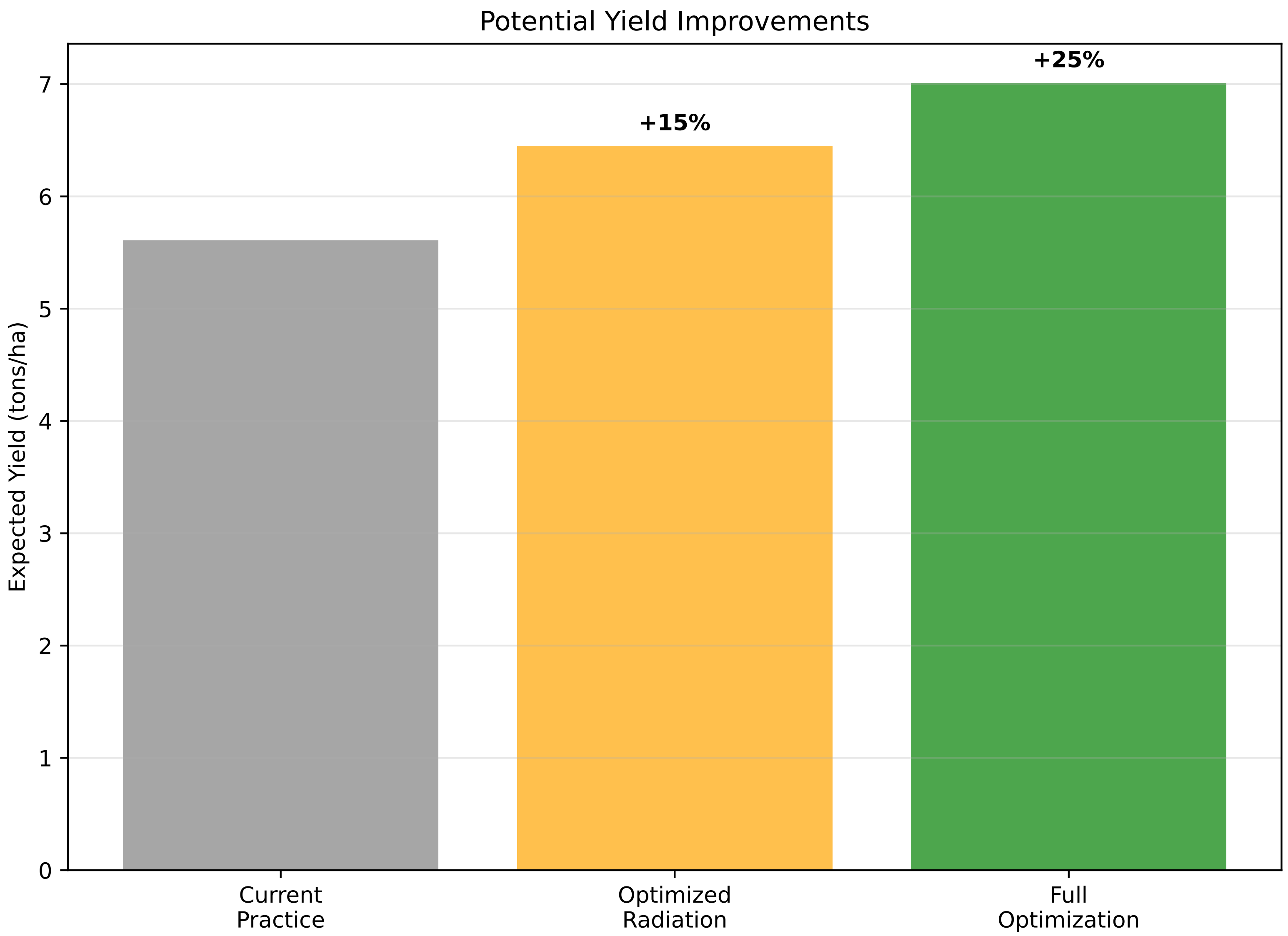
Rice Yield Optimization - Summary Dashboard

Key Research Findings

- Radiation4 dominates yield prediction (SHAP: 0.237)
- Early temperature 3× more important than late temperature
- Soil OM optimal range: 2.5-3.5% (correlation effects)
- Weather factors: 54.8% vs Soil: 32.2% importance
- Model explains 22.1% of yield variation
- RMSE: 0.163 tons/ha

Farmer Recommendations

1. Time grain filling with peak radiation season
2. Ensure 25°C+ max temps during early growth
3. Maintain soil OM 2.5-3.5% with balanced C:N
4. Apply P early when soil temps rising
5. Use AWD vegetative, flood grain filling
6. Select 110-130 day varieties for timing



Risk Assessment for Farmers

HIGH RISK: Poor radiation during grain filling

MEDIUM RISK: Low early-season temperatures

MEDIUM RISK: Soil OM outside optimal range

LOW RISK: Phosphorus deficiency (manageable)

LOW RISK: Precipitation timing (adaptable)